Chapter 1

Generative probabilistic classification: A Summary

Generative Models

- **Generative approach**: model how data in each class are produced, rather than directly separating classes.
- Each class C_k is described by a **class-conditional distribution** $p(\mathbf{x} \mid C_k)$, capturing typical variability of samples in that class.
- Generative models can be used to *sample* new observations statistically similar to training data.
- **Class priors** $p(C_k)$ express probability of observing each class before seeing data.
- Once $p(\mathbf{x} \mid C_k)$ and $p(C_k)$ are available, Bayes' rule gives **posterior** class probabilities:

$$p(C_k \mid \mathbf{x}) = \frac{p(\mathbf{x} \mid C_k)p(C_k)}{p(\mathbf{x})} = \frac{p(\mathbf{x} \mid C_k)p(C_k)}{\sum_j p(\mathbf{x} \mid C_j)p(C_j)} \tag{1.1}$$

- Parameters of $p(\mathbf{x} \mid C_k)$ and $p(C_k)$ are learned from labeled data, typically by maximum likelihood.
- Classification: select class with highest posterior probability.

Deriving Posterior Probabilities (Binary Case)

- For binary classification (C_1, C_2), Bayes' rule yields:

$$p(C_1|\mathbf{x}) = \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_1)p(C_1) + p(\mathbf{x}|C_2)p(C_2)} = \frac{1}{1 + \frac{p(\mathbf{x}|C_2)p(C_2)}{p(\mathbf{x}|C_1)p(C_1)}} \tag{1.2}$$

- Define **log odds**:

$$a(\mathbf{x}) = \log \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_2)p(C_2)} = \log \frac{p(C_1|\mathbf{x})}{p(C_2|\mathbf{x})} \tag{1.3}$$

- Log odds convert multiplicative comparisons into additive ones.
- Posterior probability expressed via **logistic (sigmoid) function**:

$$p(C_1|\mathbf{x}) = \frac{1}{1 + e^{-a(\mathbf{x})}} = \sigma(a(\mathbf{x})), \quad p(C_2|\mathbf{x}) = 1 - \sigma(a(\mathbf{x})) = \frac{1}{1 + e^{a(\mathbf{x})}} \tag{1.4}$$

- Sigmoid properties:

- Maps $\mathbb{R}$ to $(0, 1)$, producing valid probabilities.
- $\sigma(-x) = 1 - \sigma(x)$.
- Derivative: $\frac{d\sigma(x)}{dx} = \sigma(x)(1 - \sigma(x))$.

Multiclass Extension

- For $K > 2$ classes, define for each class k:

$$a_k(\mathbf{x}) = \log(p(\mathbf{x}|C_k)p(C_k)) = \log p(\mathbf{x}|C_k) + \log p(C_k) \tag{1.5}$$

- These are **unnormalized log-posteriors** (up to additive constant $-\log p(\mathbf{x})$).
- Posterior probabilities via **softmax** (normalized exponential):

$$p(C_k|\mathbf{x}) = \frac{e^{a_k}}{\sum_j e^{a_j}} = s(a_k) \tag{1.6}$$

- Softmax converts vector of real scores into probability distribution over classes.
- If one score a_k much larger than others, softmax assigns probability near 1 to class k.

Gaussian Discriminant Analysis (GDA)

Model Assumptions

- Assume Gaussian class-conditional distributions with **shared covariance matrix** $\boldsymbol{\Sigma}$:

$$p(\mathbf{x}|C_k) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}_k)\right) \tag{1.7}$$

- $\boldsymbol{\mu}_k$: center of class C_k; $\boldsymbol{\Sigma}$: common spread and correlation structure.

Binary Case

- Log odds become:

$$\begin{aligned} a(\mathbf{x}) &= \log \frac{p(\mathbf{x}|C_1)p(C_1)}{p(\mathbf{x}|C_2)p(C_2)} \\ &= \frac{1}{2}(\boldsymbol{\mu}_2^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_2 - \boldsymbol{\mu}_1^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}_1) \\ &\quad + (\boldsymbol{\mu}_1^T \boldsymbol{\Sigma}^{-1} - \boldsymbol{\mu}_2^T \boldsymbol{\Sigma}^{-1})\mathbf{x} + \log \frac{p(C_1)}{p(C_2)} \end{aligned}$$

- Quadratic terms in $\mathbf{x}$ cancel due to shared covariance $\Rightarrow a(\mathbf{x})$ is **linear** in $\mathbf{x}$.

- Therefore:

$$a(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b \tag{1.8}$$

with parameters:

$$\begin{aligned} \mathbf{w} &= \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \\ b &= \frac{1}{2}(\boldsymbol{\mu}_2^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}_1) + \log\frac{p(C_1)}{p(C_2)} \end{aligned}$$

- Posterior: $p(C_1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b)$ (generalized linear model).
- Decision boundary ($a(\mathbf{x}) = 0$) is a hyperplane:

$$\mathbf{w}^T\mathbf{x} + b = 0 \tag{1.9}$$

- With isotropic covariance $\boldsymbol{\Sigma} = \lambda\mathbf{I}$:

$$2(\boldsymbol{\mu}_2 - \boldsymbol{\mu}_1)\mathbf{x} + \|\boldsymbol{\mu}_1\|^2 - \|\boldsymbol{\mu}_2\|^2 + 2\lambda\log\frac{p(C_2)}{p(C_1)} = 0 \tag{1.10}$$

- Separation direction determined by difference of class means.

Multiclass Case ($K > 2$)

- With shared covariance, each $a_k(\mathbf{x})$ is affine:

$$a_k(\mathbf{x}) = \mathbf{w}_k^T\mathbf{x} + b_k \tag{1.11}$$

- Posterior: $p(C_k|\mathbf{x}) = s(\mathbf{w}_k^T\mathbf{x} + b_k)$ (softmax of linear functions).
- Decision boundary between classes j and k: $a_j(\mathbf{x}) = a_k(\mathbf{x})$ defines a hyperplane.
- All decision boundaries are linear; regions are convex.

General Covariance Matrices (Binary Case)

- If each class has its own covariance matrix $\boldsymbol{\Sigma}_1 \neq \boldsymbol{\Sigma}_2$, quadratic terms do not cancel.
- Log odds become:

$$\begin{aligned} a(\mathbf{x}) = &\frac{1}{2}\left((\mathbf{x} - \boldsymbol{\mu}_2)^T\boldsymbol{\Sigma}_2^{-1}(\mathbf{x} - \boldsymbol{\mu}_2) - (\mathbf{x} - \boldsymbol{\mu}_1)^T\boldsymbol{\Sigma}_1^{-1}(\mathbf{x} - \boldsymbol{\mu}_1)\right) \\ &+ \frac{1}{2}\log\frac{|\boldsymbol{\Sigma}_2|}{|\boldsymbol{\Sigma}_1|} + \log\frac{p(C_1)}{p(C_2)} \end{aligned}$$

- Decision boundary is quadratic (conic sections in low dimension).
- Classifier compares two quadratic forms; boundary surfaces are at most quadratic.

Maximum Likelihood Estimation for GDA

- Binary case, shared covariance. Parameters: $\pi = p(C_1)$, $\boldsymbol{\mu}_1$, $\boldsymbol{\mu}_2$, $\boldsymbol{\Sigma}$.
- Training set $\mathcal{T}$: $(\mathbf{x}_i, t_i)$ with $t_i \in \{0, 1\}$.
- Likelihood:

$$L(\pi, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}|\mathcal{T}) = \prod_{i=1}^{n} (\pi \cdot \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}_1, \boldsymbol{\Sigma}))^{t_i} ((1-\pi) \cdot \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}_2, \boldsymbol{\Sigma}))^{1-t_i} \tag{1.12}$$

- Log-likelihood:

$$\begin{aligned} l(\pi, \boldsymbol{\mu}_1, \boldsymbol{\mu}_2, \boldsymbol{\Sigma}|\mathcal{T}) = &\sum_{i=1}^{n} t_i \log \pi + \sum_{i=1}^{n} t_i \log \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}_1, \boldsymbol{\Sigma}) \\ &+ \sum_{i=1}^{n} (1-t_i) \log(1-\pi) + \sum_{i=1}^{n} (1-t_i) \log \mathcal{N}(\mathbf{x}_i|\boldsymbol{\mu}_2, \boldsymbol{\Sigma}) \end{aligned}$$

- MLE estimates:

$$\begin{aligned} \pi &= \frac{n_1}{n} \quad \text{(empirical class frequency)} \\ \boldsymbol{\mu}_1 &= \frac{1}{n_1} \sum_{\mathbf{x}_i \in C_1} \mathbf{x}_i \\ \boldsymbol{\mu}_2 &= \frac{1}{n_2} \sum_{\mathbf{x}_i \in C_2} \mathbf{x}_i \\ \boldsymbol{\Sigma} &= \frac{n_1}{n} \mathbf{S}_1 + \frac{n_2}{n} \mathbf{S}_2 \end{aligned}$$

  where $\mathbf{S}_k = \frac{1}{n_k} \sum_{\mathbf{x}_i \in C_k} (\mathbf{x}_i - \boldsymbol{\mu}_k)(\mathbf{x}_i - \boldsymbol{\mu}_k)^T$.
- Shared covariance is weighted average of within-class covariances.

MAP Estimation for GDA

- MAP estimation maximizes posterior $p(\theta|\mathcal{T}) \propto p(\mathcal{T}|\theta)p(\theta)$.
- Equivalent to maximizing $\log p(\mathcal{T}|\theta) + \log p(\theta)$ (log-likelihood + regularization).
- **For class prior π**: Beta prior yields posterior proportional to $\pi^{n_1+\alpha-1}(1-\pi)^{n_2+\beta-1}$; MAP estimate incorporates pseudo-counts, preventing extreme values with small data.
- **For covariance $\boldsymbol{\Sigma}$**: MAP acts as **shrinkage** estimator, improving conditioning and numerical stability. Blends empirical covariance with structured prior target (e.g., $\tau\mathbf{I}$), damping spurious correlations and extreme eigenvalues.
- **For means $\boldsymbol{\mu}_k$**: Gaussian prior shrinks empirical means toward prior center $\boldsymbol{\mu}_0$. MAP estimate is convex combination:

$$\boldsymbol{\mu}_k^{\text{MAP}} = \alpha_k \boldsymbol{\mu}_k^{\text{ML}} + (1-\alpha_k)\boldsymbol{\mu}_0, \quad \alpha_k \in (0, 1) \tag{1.13}$$

  with α_k increasing with class size n_k. Reduces variance when data limited.
- MAP improves stability and generalization, especially in high-dimensional or small-sample regimes.

Conditional Independence and Naive Bayes

- **Naive Bayes assumption**: conditioned on class label C_k, features are mutually independent:

$$p(\mathbf{x} \mid C_k) = \prod_{i=1}^{m} p(x_i \mid C_k) \tag{1.14}$$

- Features can be continuous or discrete; each $p(x_i \mid C_k)$ can be any parametric family.
- For continuous features with Gaussian marginals:

$$p(x_i \mid C_k) = \mathcal{N}(x_i \mid \mu_{ki}, \sigma_{ki}^2) \tag{1.15}$$

- This corresponds to GDA with diagonal covariance matrices (features uncorrelated given class).
- Naive Bayes drastically reduces number of parameters, beneficial for high-dimensional data, at cost of ignoring feature correlations.

Discrete Features and Language Models

Language Model Definition

- A **language model** is a categorical probability distribution over a vocabulary $\mathcal{D} = \{t_1, \dots, t_n\}$.
- Vocabulary constructed from document collection via stop word removal and stemming.
- For each term t_i, compute multiplicity m_i (total occurrences in collection).
- Probability vector $\hat{\boldsymbol{\phi}} = (\hat{\phi}_1, \dots, \hat{\phi}_n)^T$ with $\hat{\phi}_i = p(t_i|\mathcal{C})$.
- Constraints: $0 \leq \hat{\phi}_i \leq 1$, $\sum_{i=1}^{n} \hat{\phi}_i = 1$.
- Maximum likelihood estimate: $\hat{\phi}_i = \frac{m_i}{N}$ where $N = \sum m_i$ (observed frequency).

Laplace Smoothing

- MLE suffers from **zero-frequency problem**: unseen terms get probability zero.
- **Laplace smoothing** (add-one smoothing):

$$\phi_i^{Lap} = \frac{m_i + 1}{N + n} \tag{1.16}$$

- Ensures all terms have non-zero probability; distribution remains normalized.
- Bayesian interpretation: equivalent to Dirichlet prior with $\alpha_i = 1$.
- **Lidstone smoothing** (generalization):

$$\phi_i^{(\alpha)} = \frac{m_i + \alpha}{N + \alpha n}, \quad 0 < \alpha \leq 1 \tag{1.17}$$

- Smaller α gives less aggressive smoothing, preferred for large-scale NLP.

Bayesian Language Modeling (Dirichlet-Multinomial)

- Treat parameters as random variables with **Dirichlet prior**:

$$p(\phi_1, \dots, \phi_n | \boldsymbol{\alpha}) = \frac{1}{\Delta(\alpha_1, \dots, \alpha_n)} \prod_{i=1}^{n} \phi_i^{\alpha_i - 1} \tag{1.18}$$

- α_i are pseudo-counts (prior observations).
- Dirichlet is conjugate prior to multinomial.
- After observing counts m_i, posterior is $\mathrm{Dir}(\boldsymbol{\phi} | \overline{\boldsymbol{\alpha}})$ with:

$$\overline{\boldsymbol{\alpha}} = (\alpha_1 + m_1, \dots, \alpha_n + m_n) \tag{1.19}$$

- Predictive posterior (probability of next term t_j):

$$\hat{\phi}_j = \frac{\alpha_j + m_j}{\sum_{k=1}^{n} (\alpha_k + m_k)} = \frac{\alpha_j + m_j}{\alpha_0 + N} \tag{1.20}$$

  where $\alpha_0 = \sum \alpha_k$.
- This is the expected value of ϕ_j under posterior Dirichlet.
- With uniform prior $\alpha_i = \alpha$, we get $\hat{\phi}_j = \frac{m_j + \alpha}{\alpha n + N}$.
- When $\alpha = 1$, this equals Laplace smoothing.

Naive Bayes Classifiers for Text

- For document d with terms $\bar{t}_1, \dots, \bar{t}_{n_d}$, posterior probability:

$$p(C_k | d) \propto p(C_k) \prod_{j=1}^{n_d} p(\bar{t}_j | C_k) \tag{1.21}$$

- $p(C_k)$ estimated as class frequency: $p(C_k) = \frac{|C_k|}{|C_1| + |C_2|}$.
- $p(\bar{t}_j | C_k)$ are term probabilities from language model for class C_k.
- Classification rule: assign d to class with highest $p(C_k) \prod p(\bar{t}_j | C_k)$.
- Naive Bayes assumption: terms conditionally independent given class (ignores linguistic dependencies).
- Despite strong assumption, often performs well, especially with high-dimensional features.

Feature Selection via Mutual Information

- Identify terms whose presence/absence best characterizes document class.

- **Mutual Information** I_j between term t_j and class distribution:

$$
\begin{aligned}
I_j &= \sum_{k=1,2} p(t_j, C_k) \log \frac{p(t_j, C_k)}{p(t_j)p(C_k)} \\
&= p(t_j) \sum_{k=1,2} p(C_k|t_j) \log \frac{p(C_k|t_j)}{p(C_k)} \\
&= p(t_j) KL(p(C_k|t_j) \| p(C_k))
\end{aligned}
$$

- KL divergence measures how much class distribution changes given term presence.
- Weighted by $p(t_j)$ to avoid over-prioritizing rare terms.
- Estimated using language model parameters:

$$
\begin{aligned}
I_j = \phi_{j1}\pi_1 \log \frac{\phi_{j1}}{\phi_{j1}\pi_1 + \phi_{j2}\pi_2} \\
+ \phi_{j2}\pi_2 \log \frac{\phi_{j2}}{\phi_{j1}\pi_1 + \phi_{j2}\pi_2}
\end{aligned}
$$

  where $\phi_{jk} = p(t_j|C_k)$, $\pi_k = p(C_k)$.
- Rank terms by I_j, select top K for classification.

Mutual Information vs. Frequency-Based Selection

- **Frequency-based**: selects most common terms (stop words, background words) – poor for discrimination.
- **Mutual Information**: selects terms strongly correlated with specific class (high in one class, low in others).
- MI naturally ignores high-frequency words evenly distributed across classes.
- $p(t_j)$ factor prevents selection of extremely rare "noise" terms.
- MI focuses on terms where ratio $\frac{p(t_j|C_1)}{p(t_j|C_2)}$ differs significantly from 1, amplifying class separation.
- Frequency best for language modeling; MI superior for supervised classification.